Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec:Sr.Super60_NUCLEUS BT | Paper -2(Adv-2020-P2-Model) | Date: 12-10-2025

Time: 02.00Pm to 05.00Pm | GTA-03 | Max. Marks: 198

KEY SHEET

PHYSICS

1	**6**	2	**2**	3	**3**	4	**2**	5	**4**	6	**3**
7	**ABCD**	8	**ABCD**	9	**ACD**	10	**ABCD**	11	**ABCD**	12	**BC**
13	**2**	14	**17.32**	15	**8**	16	**3**	17	**1.29-1.34**	18	**34**

CHEMISTRY

19	**8**	20	**5**	21	**1**	22	**8**	23	**7**	24	**2**
25	**ACD**	26	**AB**	27	**ABCD**	28	**ABCD**	29	**AB**	30	**ABD**
31	**9**	32	**0.34-0.35**	33	**4**	34	**454.5**	35	**1250**	36	**6**

MATHEMATICS

37	**7**	38	**4**	39	**3**	40	**0**	41	**2**	42	**8**
43	**ABCD**	44	**ABC**	45	**ABCD**	46	**AD**	47	**AC**	48	**BCD**
49	**0.25**	50	**0.51**	51	**1.57**	52	**0.50**	53	**19**	54	**9.91**

For More Join: @IITJEE_Advanced

SOLUTIONS

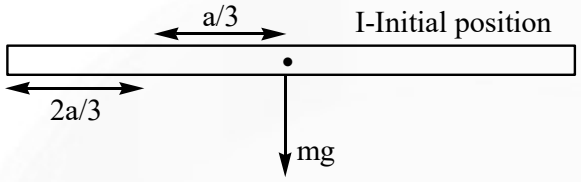
PHYSICS

1. CM will remain at rest

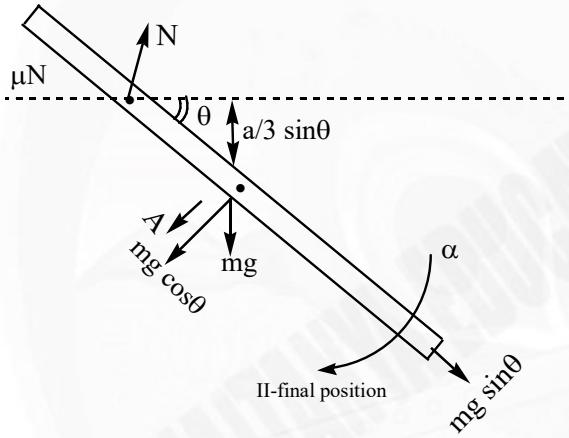
COLM $\Rightarrow mv_1 = 2mv_2$

COE $\Rightarrow \frac{1}{2}kx_0^2 = \frac{1}{2}mv_1^2 + \frac{1}{2}2mv_2^2 \qquad \Rightarrow W_B = \frac{1}{6}kx_0^2$

2. M.I of the road about the axis of rotation



$I = \frac{4ma^2}{9}$ ..... (i)



Using cons. of energy $mg\left(\frac{a}{3}\sin\theta\right) = \frac{1}{2}.\frac{4}{9}ma^2.\omega^2$

$\omega^2 = \frac{6g\sin\theta}{4a}$ ..... (ii)

suppose acceleration of COM is 'A' in the direction as shown then

$mg\cos\theta - N = mA$ ..... (iii)

for rotary motion,

$mg\cos\theta\frac{a}{3} = \frac{4}{9}ma^2\left(\frac{A}{a/3}\right) \qquad [J=I\alpha]$

$\Rightarrow \qquad mg\cos\theta = 4mA$

and $mg\cos\theta - N = mA \qquad N = 3MA$

Thus, $A = \frac{g\cos\theta}{4};\ \alpha = \frac{3g\cos\theta}{4a}$

Applying equation of circular motion on the rod in final position.

$\mu N - mg\sin\theta = m\omega^2.\frac{a}{3} \qquad \mu\frac{3mg\cos\theta}{4} - mg\sin\theta = m\frac{6g\cos\theta}{4a}.\frac{a}{3}$

$\Rightarrow 3\mu g\cos\theta - 4g\sin\theta = 2g\sin\theta \qquad 6g\sin\theta = 3\mu g\cos\theta$

$\mu = 2\tan\theta$

3. The initial state of the system is shown in centre of mass frame. The block A moves towards left and block B moves towards right. Both of them travel by x1 and x2

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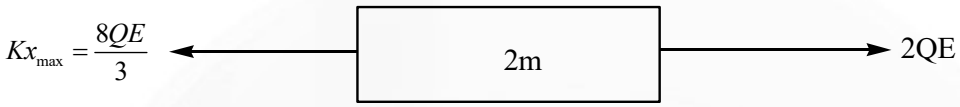
respectively before coming to rest momentarily at same time. The length of spring is maximum at the stage.

Applying work energy theorem we get

$$\frac{4QE}{3}x_1+\frac{4QE}{3}x_2=\frac{1}{2}k(x_1+x_2)^2$$

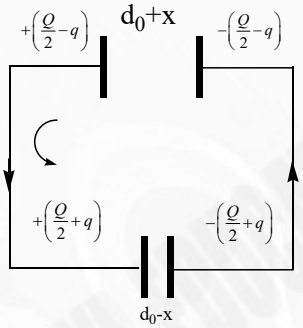
or the maximum extension is $(x_1+x_2)=\frac{8QE}{3k}$

In the state of maximum extension the FBD of B in ground frame is



$\therefore$ Acceleration of B is $\frac{\frac{8QE}{3}-2QE}{2m}=\frac{QE}{3m}$

4. Let each plate moves a distance x from its initial position.



Let q charge flows in the loop. Using KVL

$$\frac{\left(\frac{Q}{2}-q\right)(d+x)}{\varepsilon_0 A}-\frac{\left(\frac{Q}{2}+q\right)(d-x)}{\varepsilon_0 A}=0$$

$\therefore q=\frac{Qx}{2d_0}$; $I=\frac{dq}{dt}$ $=\frac{Q}{2d_0}\left(\frac{dx}{dt}\right)$; $I=\frac{Qv_0}{2d_0}$

5. $\because$ CM of two body system lies along Y-axis, the x-coordinate of center of mass of two body system is $X_{CM}=0$

$\therefore$ 3m × x = m × 9 $\Rightarrow$ x=3

$\therefore$ Length of string = $\sqrt{5^2+(9+3)^2}$ =13 $=x^2-x+1$ (given) $\Rightarrow$ x=4

6. The ball experiences centrifugal force F_C and electric force F_e both in radially outward direction.

$$\Rightarrow F=F_c+F_e=m\omega^2x+q.\left(\frac{KQx}{R^3}\right) \quad \Rightarrow V_r\frac{dV_r}{dx}=\left(\omega^2+\frac{kQq}{mR^3}\right)x$$

Where V_r=radial velocity.

Integrating both sides, we get $\Rightarrow V_r=\sqrt{2}\omega R$

At point of leaving. the ball will also be having tangential velocity

$V_t=\omega R$

From Eqs (i) and (ii), we get $V=\sqrt{V_r^2+V_t^2}=\sqrt{3}\ \omega R$

7. Motion of P relative to centeroid of the triangle

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$-\frac{dr}{dt} = V\cos 30^0$ ..... (1)

$\frac{rd\theta}{dt} = V\sin 30^0$ ..... (2)

$\frac{(1)}{(2)} \Rightarrow r = \frac{1}{\sqrt{3}} e^{-\sqrt{3}\,\theta}$

After one revolution around center $\theta=2\pi$ at $t=0, \theta=0$

Time taken to meet $= \frac{l}{V_{rd}}$, approach $= \frac{2l}{3V}$

$\therefore$ Average velocity $V = \frac{l/\sqrt{3}}{\frac{2l}{3V}} = \frac{\sqrt{3}V}{2}$ $\quad t = \frac{2l}{3V}\left(1 - e^{-2\pi}\right)$

Radius of curvature $R_c = \frac{V^2}{a_\perp}$ $\quad a_\perp = V\omega$

Radius of curvature $R_c = \frac{2l}{\sqrt{3}}$

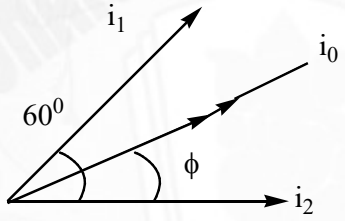
8. In RC Branch $Z_1 = \sqrt{\left(\sqrt{3}R\right)^2 + R^2}$

$= 2R$

Current leads voltage in phase by $\tan\phi = \frac{X_c}{R} = \frac{\sqrt{3}R}{R}$

$\Rightarrow \phi_1 = 60^0,\ i_1 = \frac{V_0}{2R}$

In LCR branch $Z_1 = 2R,\ \phi_1 = 0,\ i_2 = \frac{V_0}{2R}$



$i_0 = \frac{\sqrt{3}\,V_0}{2r}$ $\quad \phi = \frac{\pi}{6}$ $i_0 = \left(\frac{\sqrt{3}\,V_0}{2R}\right)\sin(\omega t + \pi/6)$

Effective impedance$= \frac{2R}{\sqrt{3}}$ $\quad \left(Z_{eff} = \frac{V_0}{i_0}\right)$

Effective reactance$= Z\sin\phi$ $\quad \frac{2R}{\sqrt{3}}\frac{1}{2} = \frac{R}{\sqrt{3}}$

Effective wattless current$= i_{rms}\sin\phi$

9. (A) Field id directed towards z-axis

(B)All points have +ve value of potential

(C) $\frac{mv^2}{A} = Q|E| = QA$

(D) V=constant$= x^2 + y^2$ i.e. (equation of cylinder)

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10. Conceptual

11. Conceptual

12. Conceptual

13. $q_A = \frac{-Q_0}{2}$ at t=0

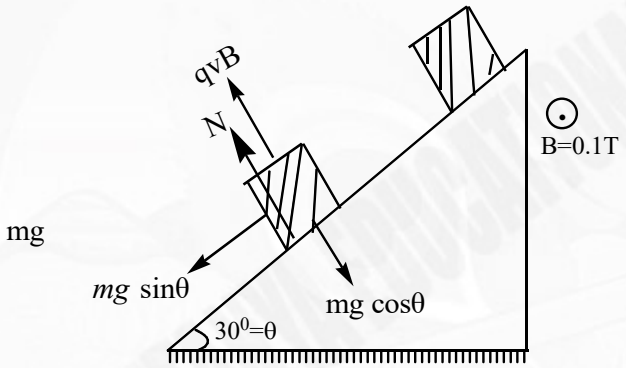
KVL $\Rightarrow$ $R\frac{dq}{dt} = E - \frac{q}{c}$

$$\int_{\frac{-Q_0}{2}}^{\frac{+Q_0}{4}} \frac{dq}{E-\frac{q}{c}} = \frac{1}{R}\int_{t=0}^{t} dt \quad \Rightarrow \ln\left(E-\frac{q}{c}\right)\Big|_{\frac{-Q_0}{2}}^{\frac{+Q_0}{4}} = -\frac{1}{Rc}t \Rightarrow E - \frac{Q_0}{4c} = \left(E+\frac{Q_0}{2c}\right)e^{-t/Rc}$$

$$\left(E-\frac{E}{4}\right) = \left(E+\frac{CE}{2c}\right)e^{-t/Rc} \qquad \frac{3E}{4} = \frac{3E}{2}e^{-t/Rc} \Rightarrow t = Rc\ln(2)$$

14. The FBD of the block is as shown below

At time t, Acceleration $= \frac{mg\ \sin\theta}{m} = g\sin\theta$



$F_{magnetic} = qvB$ $\Rightarrow$ Velocity, $v = u+at = g\sin\theta t$

$\Rightarrow$ Magnetic force shown $= qvB$ $= q\ g\sin\theta Bt$

For leaving contact, it should just exceed $mg\cos\theta$ $(N=0)$

$$\Rightarrow q\ g\sin\theta Bt = mg\cos\theta \quad \Rightarrow \quad t = \frac{m\cot\theta}{qB} = 10\sqrt{3} = 17.32s$$

15. First we have to find a point where the resultant field due to both is zero. Let the point P be at a distance x from centre of bigger star.

$$\Rightarrow \frac{G(16M)}{x^2} = \frac{GM}{(10a-x)^2} \qquad \Rightarrow x = 8a \qquad (\text{from } O_1)$$

I.e. Once the body reaches *P*, the gravitational pull of attraction due to *M* takes the lead to make *m* move
towards it automatically. So, a minimum *KE* or velocity has to be imparted to m from surface of 16 M such that it is just able to cross *P*. By law of conservation of energy.
Total mechanical energy at surface of bigger star =Total mechanical energy at *P*

$$\Rightarrow k_{\min} + \left[\frac{G(16M)m}{2a} - \frac{GMm}{8a}\right]$$

$$= 0 + \left[\frac{GMm}{2a} - \frac{G(16M)m}{8a}\right] \quad \Rightarrow |k_{\min}| = \frac{GMm}{8a}(45)$$

Comparing with given values, we get $k = 8$

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16. For $L-C_1$ system, $\omega_1=\frac{1}{\sqrt{LC_1}}=\frac{100}{3}\text{rad/s}\Rightarrow T_1=2\pi\sqrt{LC_1}=\frac{6\pi}{100}\text{s}$

For $L-C_2$ $\omega_2=\frac{1}{\sqrt{LC_2}}=100\,\text{rad/s}$ $\Rightarrow T_2=2\pi\sqrt{LC_2}=\frac{2\pi}{100}\text{s}$

Circuit $L-C_1$ is ON for $\frac{1.5\pi}{100}=\frac{T_1}{4}\text{s}$

The charge on capacitor C_1 will be zero. So, the entire energy $\left(\frac{1}{2}C_1V_1^2\right)$ will go in inductor. Now, after $\frac{1.5\pi}{100}=\frac{T_2}{4}\text{s}$, current in the inductor will be zero. So, now all the energy $\left(\frac{1}{2}C_1V_1^2\right)$ will go in the capacitor C_2.

$\Rightarrow\frac{1}{2}C_1V_1^2=\frac{1}{2}C_2V_2^2\Rightarrow\frac{1}{2}(900\mu F)(1)^2=\frac{1}{2}(100\mu F)V_2^2\Rightarrow V_2=3\text{V}$

17. ${}_1H^2+{}_1H^2\rightarrow{}_1H^3+p$

${}_1H^2+{}_1H^3\rightarrow{}_2He^4+n$

Net Reaction $3\,{}_1H^2\rightarrow{}_2He^4+p+n$

$\Delta m=3(2.014)-[4.001+1.007+1.008]=0.026$

3 deuterons release $3.87\times10^{-12}\text{J}$

$\therefore$ 10th deuterons release $=\frac{3.87\times10^{-12}\times10^{40}}{3}=1.29\times10^{28}\text{J}$

$P=\frac{E}{t}\Rightarrow t=\frac{E}{P}=\frac{1.29\times10^{28}}{10^{16}}=1.29\times10^{12}=1.29\text{Ts}$

18. Let e in hydrogen atom is exited to nth level.

$\therefore E_{KE(n-1)}=8\left|E_{p.E(n)}\right|$ $\therefore 13.6\,eV=8\left|2\times\frac{13.6}{n^2}ev\right|$ $\Rightarrow$ n=4

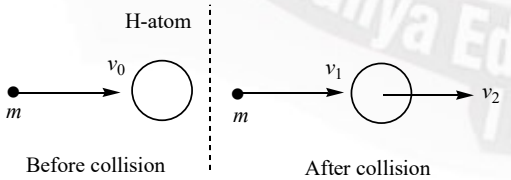
$\therefore\Delta E=13.6\left(1-\frac{1}{16}\right)=\frac{15}{16}\times13.6\,eV=12.75\text{ eV}$

Using of conservation of linear momentum

$\text{mv}=\text{mv}_1+mv_2$ .....(1)

$(\text{v}_2-v_1)=\frac{1}{2}v$ .....(2)

$\therefore\text{v}_1=\frac{v}{4},v_2=\frac{3v}{4}$ .....(3)



By energy conservation

$\therefore\frac{1}{2}mv^2=\frac{1}{2}mv_1^2+\frac{1}{2}mv_2^2+\Delta E$ $\Rightarrow\frac{1}{2}mv^2\left(1-\frac{1}{16}-\frac{9}{16}\right)=\Delta E$

$\Rightarrow\frac{1}{2}mv^2=\frac{16}{6}\Delta E=\frac{8}{3}\times12.75eV=34eV$

K.E of neutron=$34eV$

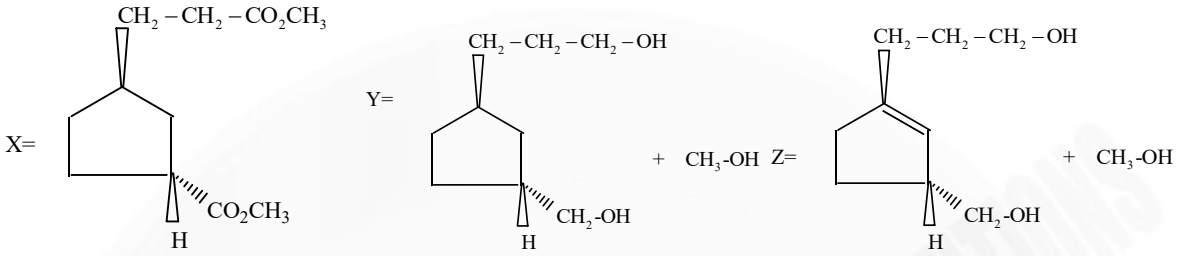
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CHEMISTRY

19. No. of valence shell electrons in X=4, Y=3, Z=1

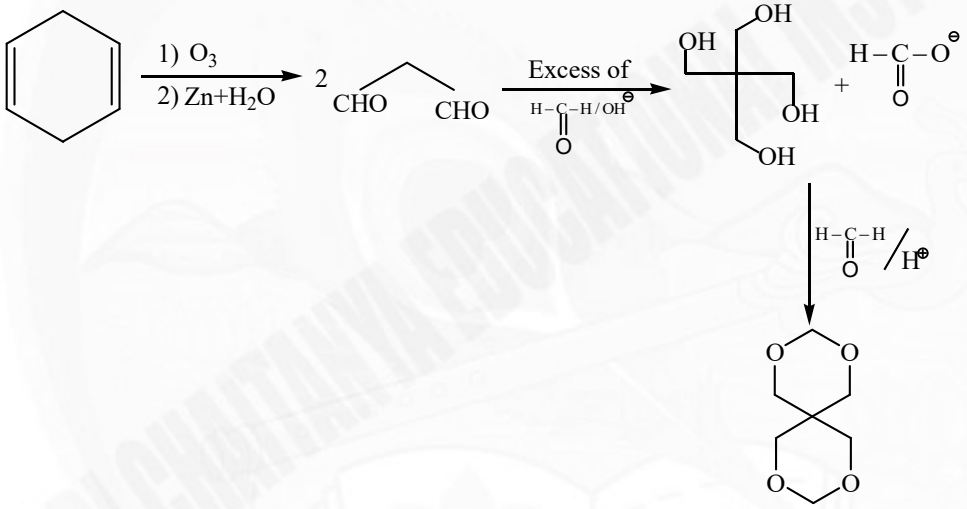
20. $C_2H_2, C_2H_4, C_2H_2Cl_2 trans, B_3N_3H_6, C_6H_6$. These are non- polar and having nodal plane in the molecular plane only.

21.

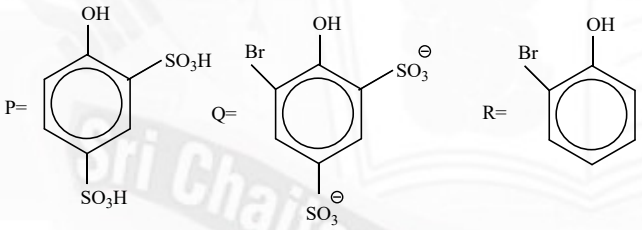


22. $2KMnO_4 + 16\,HCl \rightarrow 2KCl + 2MnCl_2 + + 8H_2O + 5Cl_2$

23.



24.



25. **Paramagnetic compound (X)** are attracted towards magnetic field and the pan is deflected downwards.
While the **Diamagnetic compound (X)** are repelled by magnetic field and pan is deflected upward.

A) $X \Rightarrow H_2O_2 \rightarrow$ **Diamagnetic** (correct)

B $X \Rightarrow K_3\left[Fe(CN)_6\right](s) \rightarrow$ **Paramagnetic**

(incorrect) Here Fe^{3+} + Strong field ligand $\rightarrow$

$3d^5 \Rightarrow \left[t_2g^5, eg^0\right]$

C) $X \Rightarrow O_2 \rightarrow$ **Paramagnetic** (correct)

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Here $O_2(g)$ is paramagnetic due to two-unpaired electrons present in π^* (antibonding orbitals).

D) X $\Rightarrow$ $B_2 \rightarrow$ **Paramagnetic** (correct) It is due to presence of 2 unpaired electrons.

26. 1^0 RX Undergoes SN^2 reaction (2nd order)

For 2nd order reaction:- $t_{1/2} \alpha \frac{1}{[A]_0^{2-1}}$ $r_0 = k[P]_0^2$

27. A) $2Na[Al(OH)_4]_{(aq.)} + CO_2 \rightarrow Na_2CO_3 + H_2O + 2Al(OH)_3(\downarrow)$(or) $Al_2O_3.2H_2O(ppt)$

B) Addition of Na_3AlF_6 lowers the melting point of alumina.

C) $Graphite + O_2 \rightarrow CO + CO_2$

D) In electrorefining of aluminium the layer of pure Al act as cathode

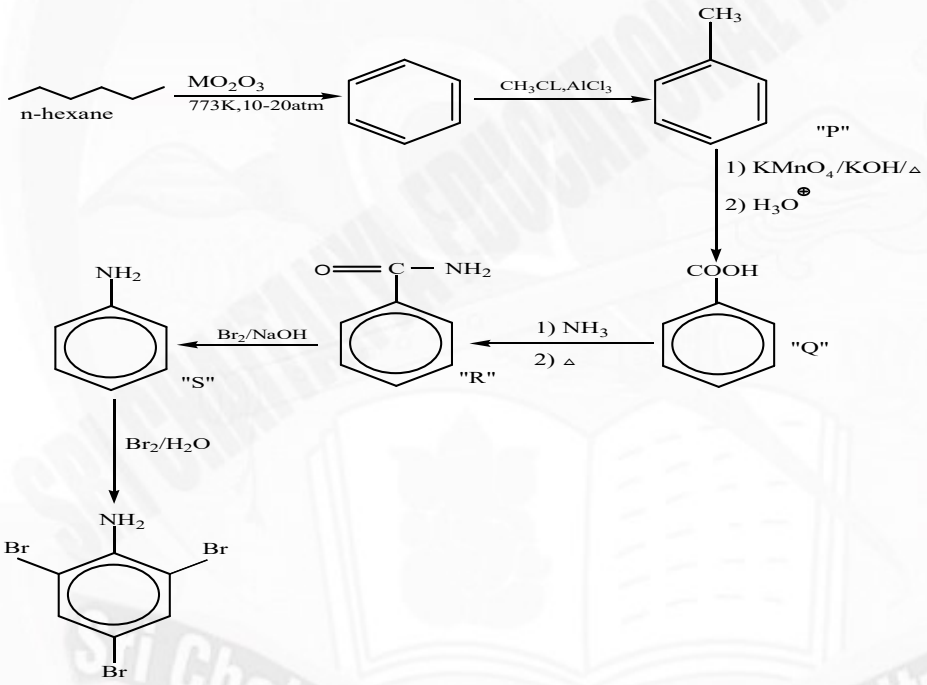
28. A) $SnCl_2.2H_2O$ is a reducing agent since Sn^{2+} tends to convert into Sn^{4+}.

B) $SnO_2 + 2KOH + 2H_2O \rightarrow K_2[Sn(OH)_6]$

C) SnI_4 orange colour due to LMCT

D) The brown colour precipitate of SnS soluble in yellow ammonium sulphide.

29.



30. E_2 & E_1 reactivity order is 3^0R-X>2^0R-X>1^0R-X

SN^1 reactivity order is 3^0R-X>2^0R-X>1^0R-X

SN^2 reactivity order is 3^0R-X<2^0R-X<1^0R-X

31. For 2nd order reaction:- $t_{1/2} \alpha \frac{1}{[A]_0^{2-1}}$ $r_0 = k[P]_0^2$ $(pK_b)_{A^-}$ =14-5=9

32. $P_T = P_{A^0}X_A + P_{B^0}X_B$

$0.5 = P_{A^0} \times 0.75 + P_{B^0} \times 0.25$ .....(1)

$0.4 = P_{A^0} \times 0.50 + P_{B^0} \times 0.50$ .....(2)

On solving $P_{A^0} = 0.6$ & $P_{B^0} = 0.2$

The pressure at which $y_A - x_A$ has maximum $= \sqrt{P_{A^0}.P_{B^0}} = \sqrt{0.6 \times 0.2} = 0.35$

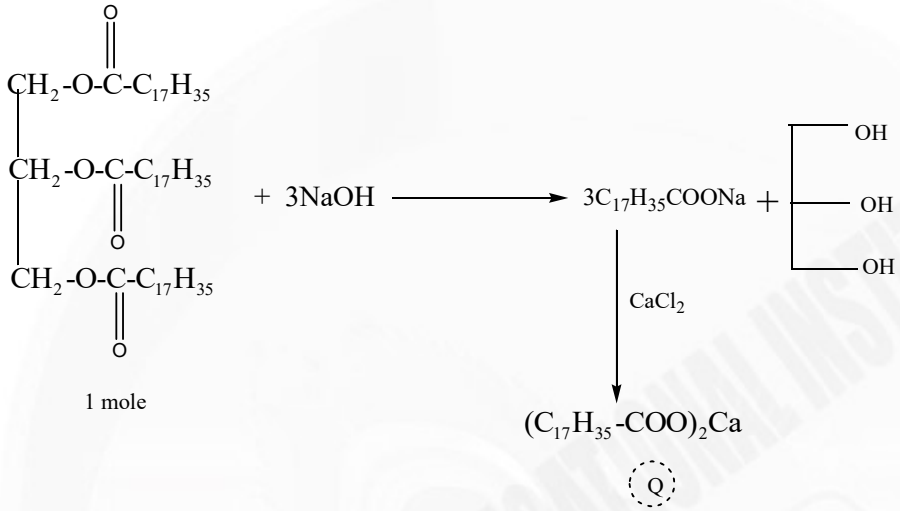
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33. $n\lambda = 2\pi r_n \qquad n \times 2.49 = 2 \times 3.14 \times 0.53 \times \frac{n^2}{Z}$

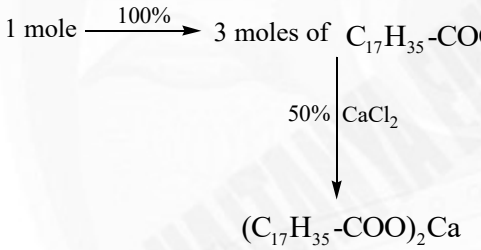
$$Z = \frac{2 \times 3.14 \times 0.53 \times n}{2.49} = 1.33\, n \quad \text{.....(1)}$$

$$193.42 = 13.6 \times Z^2 \left(\frac{1}{1^2} - \frac{1}{n^2} \right) \quad \text{.....(2)} \qquad \text{From (1) \& (2)} \quad Z=4$$

34.



1 mole



0.75 moles of $(C_{17}H_{35}\text{-}COO)_2Ca$

$606 \times 0.75 = 454.5\text{g}$

35.

	$ZnO(s)$	+ C(s)	→ 2Zn(s)	+ CO(g)
$\Delta_f H^0$	-350	0	0	-110
$\Delta_f S^0$	44	6	42	200

$\Delta_r H^0 = (-110) - (-350) = 240\ \text{kJ} \qquad \Delta_r S^0 = [42 + 200] - [44 + 6] = 192\ {}^{J}/_{K}$

$$T = \frac{\Delta H}{\Delta S} = \frac{240 \times 10^3}{192} = 1250K$$

36. $Al^{+3}(aq) + 4OH^-(aq) \rightleftharpoons Al(OH)_4^-(aq);\ 10^{34} = \dfrac{10^{-3}}{\left[Al^{+3}\right]\left[OH^-\right]^4}$ ..... (1)

$Al(OH)_3(s) \rightleftharpoons Al^{+3}(aq) + 3OH^-(aq);\ 10^{-33} = \left[Al^{+3}\right]\left[OH^-\right]^3$ .....(2)

(1) & (2), will give

$10 = \dfrac{10^{-3}}{\left[OH^-\right]} \qquad \left[OH^-\right] = 10^{-4}\,\text{M}\ \ P^{OH} = 4, \text{pH} = 10 = x$

Putting value of $\left[Al^{+3}\right]$ in eq^n (2)

$10^{-33} = 10^{-3}\left[OH^-\right]^3 \quad 10^{-30} = \left[OH^-\right]^3 \left[OH^-\right] = 10^{-10}\ \ P^{OH} = 10\ \text{pH} = 4 = y\ \ \text{x-y} = 10\text{-}4 = 6$

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MATHEMATICS

37. Let pt M has y-coordinate= $a\cos\theta$ $\therefore x = a\ln(\sec\theta+\tan\theta) - a\sin\theta$

Hence S: $x^2 + y^2 = a^2$ $\therefore A_{DC} = \pi(\sqrt{2}a)^2 = 2\pi a^2$

38. $A_1 = \int_0^a (\sin x - f(x))\,dx$ = 1-sina+(a-1)cosa

NL rule $\rightarrow f(x) = x\sin x$

$\therefore A_1 = 1-\sin 1$ $\therefore A_2 = \pi - 1 - \sin 1$ $\therefore A_3 = 3\pi - 2$

39. Upon solving AQ $\rightarrow 2(2r) = 4r$

40. Let $x = 2025^t$ $\therefore \sqrt{2025}\,.\,2025^{t^2} = 2025^{2t}$ $\therefore t^2 - 2t + \frac{1}{2} = 0$

Now $x_1\ x_2 = 2025^{t_1+t_2} = 2025^2$ $\therefore k = 0$

41. $p = \frac{48+48+1+4}{{}^{52}C_2} = \frac{101}{{}^{52}C_2}$

42. Let $\vec{u} = \frac{r}{2}\left(\hat{i} + \sqrt{3}\hat{j}\right)$

Also $\left|\vec{u} - \hat{i}\right|^2 = \left|\vec{u}\right|\left|\vec{u} - 2\hat{i}\right|$

$\therefore r = \sqrt{2} - \sqrt{1}$ $\therefore$ a=2 & b=1

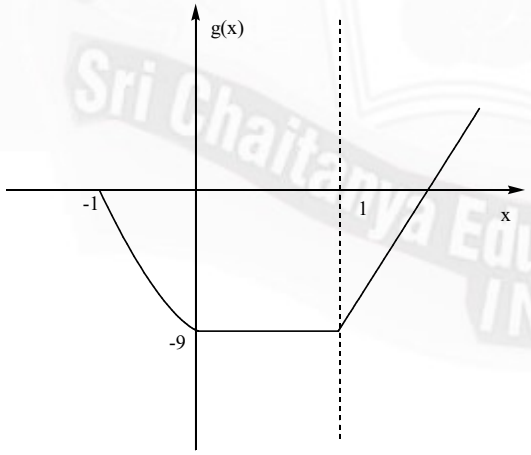
43. A) $f'(\alpha) = \frac{1-0}{\frac{\pi}{2}-0} = \frac{2}{\pi}$ (LMVT)

B) Let $g(x) = (f(x))^2 \rightarrow$ (LMVT)

C) Let $g(x) = \frac{\pi^2}{4}f(x) - x^2 \rightarrow$ Rolle's theorem

D) Let $g(x) = \sin^{-1}(f(x)) \rightarrow$ (LMVT)

44. $g(x) = \begin{cases} f(x+1) ; & -1 \le x \le 0 \\ -9 ; & 0 < x \le 1 \\ x-10 ; & 1 < x < \infty \end{cases}$



45. Hence m= $\frac{n(n-1)}{2}$

Max fresh lines = ${}^{m}C_2 - n.{}^{n-1}C_2$

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$$= \frac{n(n-1)(n-2)(n-3)}{8}$$

46. (A, B) Assume points as $\left(\alpha,\frac{1}{\alpha}\right);\left(\beta,\frac{1}{\beta}\right);(x,y)$

(C, D) Assume points as $(cos\alpha,\sin\alpha),(cos\beta,\sin\beta),(Cos\gamma,\sin\gamma)$ & (x,y)

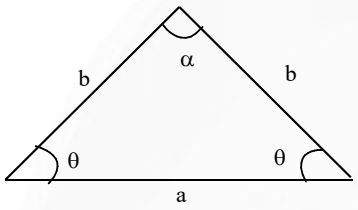
47. Conceptual

48. Apply $R_3 \to R_3 - R_2$; $R_2 \to R_2 - R_1$ $\Rightarrow f(x) = -2d^2$

(Where d is C.D of AP)

$\therefore d^2 = 1 \Rightarrow d = \pm 1 \Rightarrow d_1 = -1$ & $d_2 = 1$ & $g(x) = x^2 - 1$

49.



$$\frac{a}{\sin\alpha} = \frac{b}{\sin\theta} = 2R$$

Also $\alpha = \pi - 2\theta < \frac{\pi}{2}$

$\therefore \theta \in \left(\frac{\pi}{4},\frac{\pi}{2}\right)$

$$\therefore R = \frac{L}{4\sin\theta(1+\cos\theta)}$$

$$\therefore R_{Min} = \frac{L}{4}$$

50. $\alpha + \beta = 3 + 4i$ & $\alpha\beta = 13 - i$

Also $b = \left(\alpha\beta + \overline{\alpha\beta}\right) + \left(\alpha+\beta\right)\left(\overline{\alpha+\beta}\right) = 51$

51. Assume pt as (h, k)

$\therefore$ Chord of contact of (h, k) to $x^2 + 2y^2 = 6$ touches $x^2 + 4y^2 = 4$

Upon solving $h^2 + k^2 = 9$ (Director circle)

52. $N: y + xt = 2at + at^3$; t_1, t_2, t_3

Passes through (h, k)

$k + ht = 2at + at^3$; t_1, t_2, t_3

Also $S: x^2 + y^2 + 2gx + 2fy + c = 0$

Passes through $\left(at^2,\ 2at\right)$

$\therefore S:\ 2x^2 + 2y^2 - 2(h+2a)x - ky = 0$

53. Upon solving : Radius of $C_3 = 14$

$$\therefore l = 2\sqrt{14^2 - 6^2} = \frac{8\sqrt{10}}{1} \qquad \therefore m = 8 ;\ n = 10;\ p = 1$$

54. Use $[x] \le x < [x] + 1$

$\therefore N = (1 + 2 + \ldots\ldots\ldots + 44) + 1$

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